Run report: run-20261012T0002Z-binary-validate

What ran. `./target/release/examples/binary_gls_validate --m 31 --count 3 --hmax 320 --lmax 61 --dmax 500000 --points 24 --seed 20261012 --json research/isogeny_loop_glv_20261012/binary_validate.json`. Final source (clippy-clean). rustc 1.97.0, release, cloud container shared with another agent's builds, not isolated. Start: `../../binary_validate_start_utc.txt`. Wall time: `../../binary_validate.log`. The run id is the start of the first attempt (00:02Z); the recorded attempt 4 started later, and its start time is in that file.

Method. Curves over $F_2{}^{31}$ are built from H_D mod 2, with H_D computed natively; the field is GF(2^{62}). The walk takes an odd ℓ-isogeny from Kohel kernel polynomials, keeps it only if its codomain is on the crater, and closes it with F^{kk}. The GLS twist is $a_2 + \delta$ with $\mathrm{Tr}(\delta) = 1$. Checks run x-only on 24 points of prime order r on E′:

- $x(\psi P) = x^q$ against $x([\mu]P)$;
- $(R(x))^{2kk}$ against $x([\lambda(\pm s)]P)$.

Runs requested / completed. Four attempts; one recorded.

1. hmax 160, $\ell \le 41$, $|D| < 10^7$: 1 discriminant, PASS. Kept as `../../binary_validate_run1_hmax160.*`.
2. hmax 320, $\ell \le 61$, $|D| < 10^7$: printed the 2 PASS rows below, then was OOM-killed at 11.5 GB resident (kernel log). No JSON.
3. Same parameters: the same 2 PASS rows, then OOM-killed at 13.9 GB after 11 min 50 s. Kept as `../../binary_validate_attempt3_killed.*`.
4. `--dmax 500000` added, which stops the search past both discriminants: the run recorded here.

Which stage exhausts memory on the later candidate was not isolated. Commands and kernel log lines: `../../binary_validate_cmd.txt`.

Recorded values (`../../binary_validate.md`):

D	h	loop	H_D bits / worst err	ψ	loop acts as
−39583	62	ℓ = 31, 11 F	4162 / 2^{-200}	24/24	±λ(s) 24/24
−482623	186	ℓ = 31, 12 F	14081 / 2^{-473}	24/24	±λ(s) 24/24

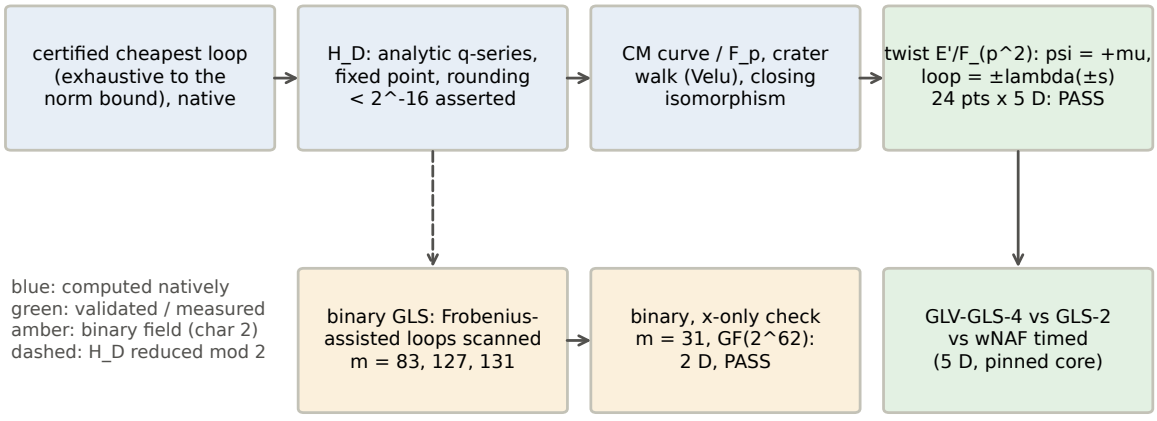


Figure 1: Where this check sits (amber, bottom middle). Diagram, not data: two pass/fail rows give no comparable quantities to plot.

Observations only. Interpretation: `../../README.md` §4.